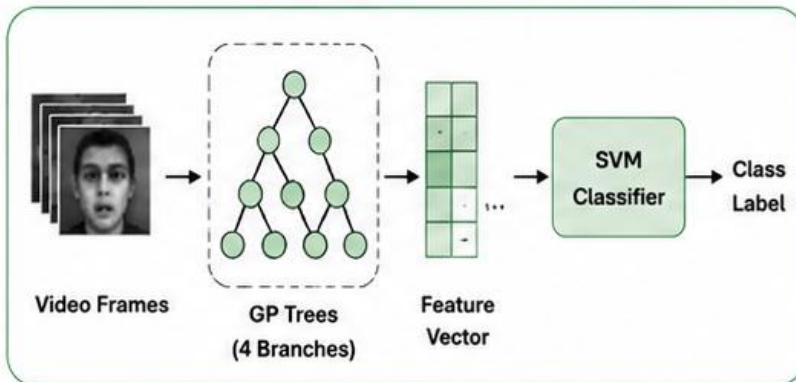


# Comparative Analysis of Genetic Programming and Large Language Model-Based Feature Extraction

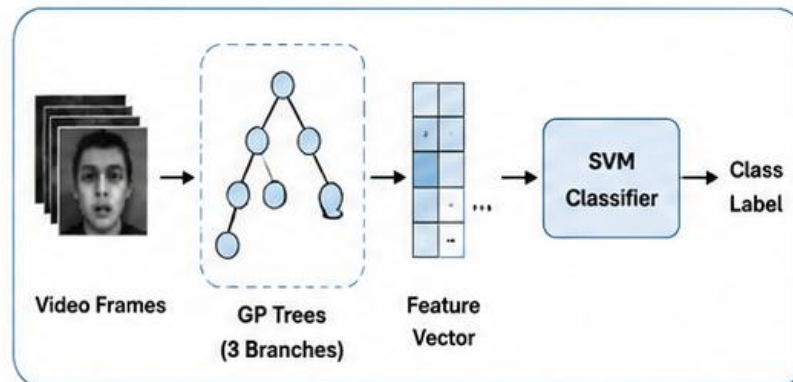
MARS Research Project

James Davies

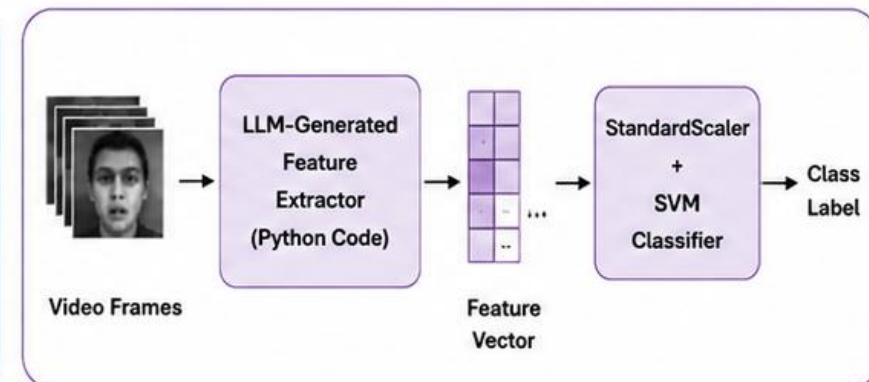
Original GP



Modified GP



LLM (EOH)



# Issue 1: Investigating Overfitting in the LLM-Generated Feature Extractor

## OBSERVATIONS

- Training accuracy is consistently **higher** than testing accuracy.
- The LLM-generated feature extractor produces a **761-dimensional** feature vector.
- High feature dimensionality increases the **risk of fitting noise** in the training data.
- Generalisation performance is **reduced** on unseen test samples.

## CONCLUSION

The LLM-generated feature extractor demonstrates signs of overfitting. The model learns patterns specific to the training data and does not generalise effectively to unseen data.

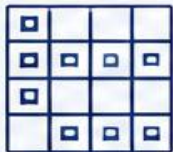
**Future work should focus on reducing feature complexity and improving generalisation.**

### TRAINING vs TESTING PERFORMANCE



Average gap between training and testing accuracy:  
**27.9%**

### FEATURE DIMENSIONALITY



**761** features generated

A high-dimensional feature space increases model complexity and the likelihood of overfitting.





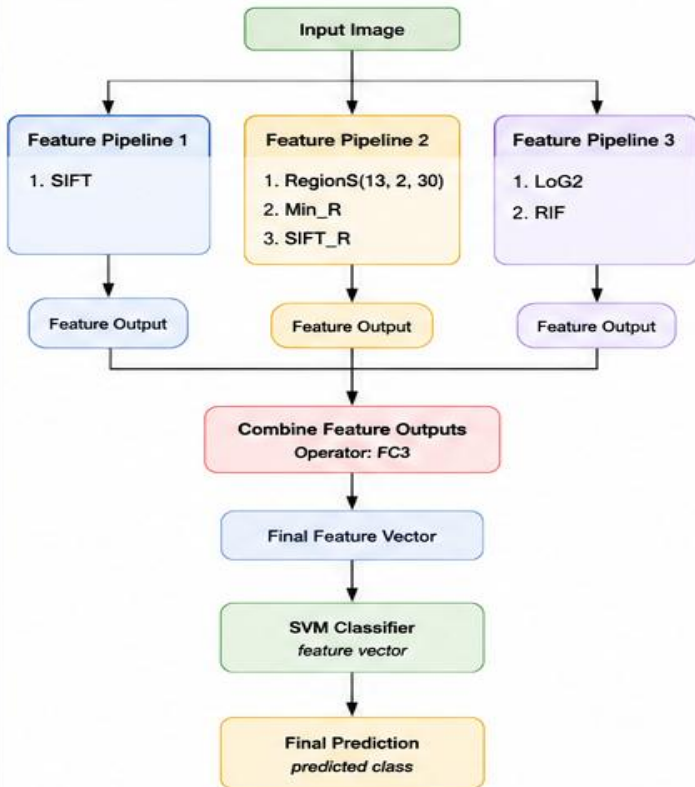
# Issue 2: Visualisation of the Feature Extraction Pipelines



Operational pipeline diagrams were developed for all three approaches to make the feature extraction process **transparent** and enable **direct comparison** between **GP (original & modified)** and the **LLM-generated (EOH)** approach.

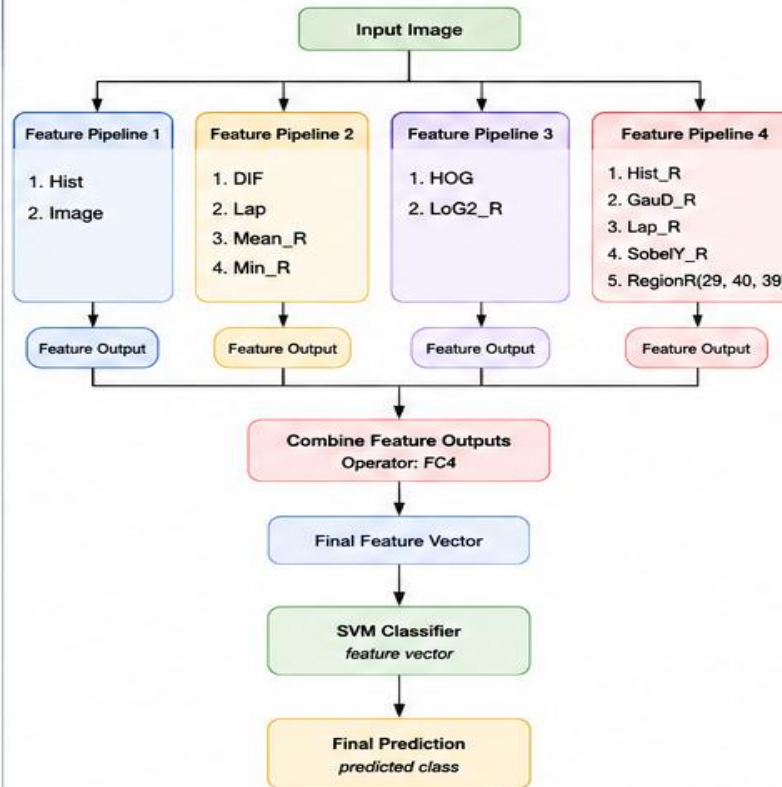
## 1. Original GP Pipeline (Best GP Programme)

Seed/run: 5 | Accuracy: 100.00



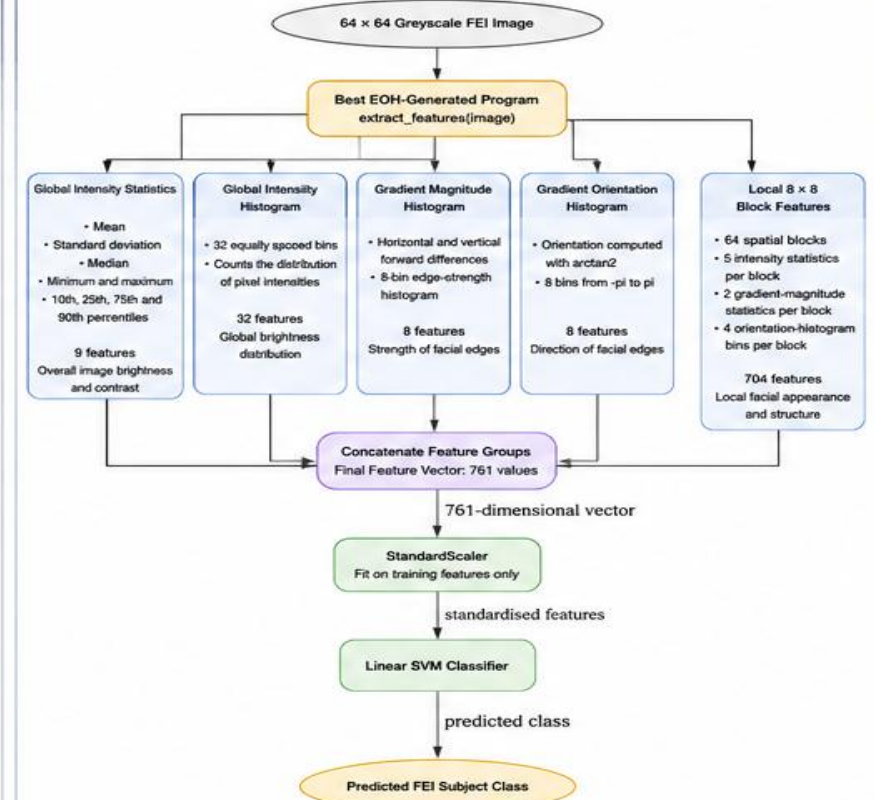
## 2. Modified GP Pipeline (Best GP Programme)

Seed/run: 3 | Accuracy: 100.00



## 3. LLM-Generated (EOH) Pipeline (Best EOH Programme)

Operational pipeline of the best EOH-generated FEI feature extractor



**Key Takeaway:** These visualisations clarify how each approach transforms raw images into discriminative feature vectors. They highlight the structured nature of GP pipelines and enable a direct comparison with the LLM-generated (EOH) methodology.



# Issue 3: Lower Average Performance of Modified GP



**Observation:** The modified GP pipeline achieved a **lower average accuracy** compared to the original GP pipeline, despite using a similar evolutionary framework.

## KEY OBSERVATIONS



Original GP achieved a **higher mean accuracy** across 10 runs.



Modified GP shows **greater variability** in performance.

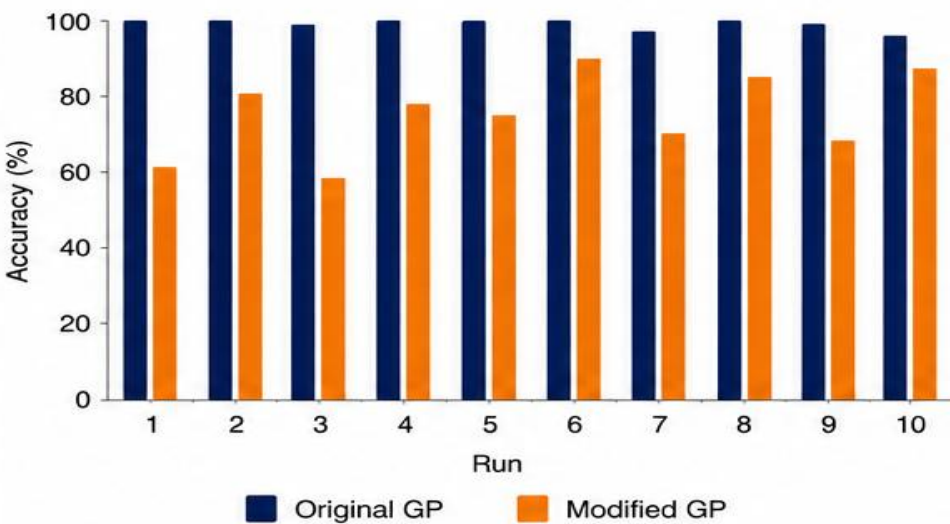


Standard deviation is **higher** for Modified GP.



This suggests that the modifications introduced may have impacted **consistency** and overall **stability**.

## ACCURACY COMPARISON ACROSS 10 RUNS



| Metric                 | Original GP | Modified GP |
|------------------------|-------------|-------------|
| Mean Accuracy (%)      | 96.90       | 76.80       |
| Standard Deviation (%) | 2.59        | 11.41       |
| Median Accuracy (%)    | 98.80       | 77.50       |
| Best Run Accuracy (%)  | 100.00      | 92.00       |
| Worst Run Accuracy (%) | 93.80       | 58.00       |

## POSSIBLE REASONS FOR LOWER AVERAGE PERFORMANCE IN MODIFIED GP



Changes to operators or parameters may have reduced **search efficiency**.



Altered representation could limit the **diversity** of candidate solutions.



Fitness landscape may be more complex, **slowing convergence**.



Trade-off between **exploration** and **exploitation**.



**CONCLUSION:** The modified GP pipeline achieves a **lower average accuracy** and **higher variability** than the original GP pipeline. Further investigation is required to identify which modifications contributed most to this performance gap and how they can be refined.



# 5. Explainability Analysis of the LLM-Generated Feature Extractor

## Key Explainable Feature Groups



### 1. Global Intensity Statistics

- Mean, standard deviation, median, minimum, maximum and percentiles.
- Capture overall brightness and contrast.



### 2. Global Intensity Histogram

- 32-bin histogram of pixel intensities.
- Summarises the distribution of intensities across the whole image.



### 3. Gradient-Based Features

- Horizontal & vertical gradients.
- Gradient magnitude and orientation.
- Histograms of edge strength and edge direction.



### 4. Local Block-Based Statistics

- Image divided into an  $8 \times 8$  grid.
- Intensity, gradient magnitude and orientation statistics per block.
- Preserves local facial structure.

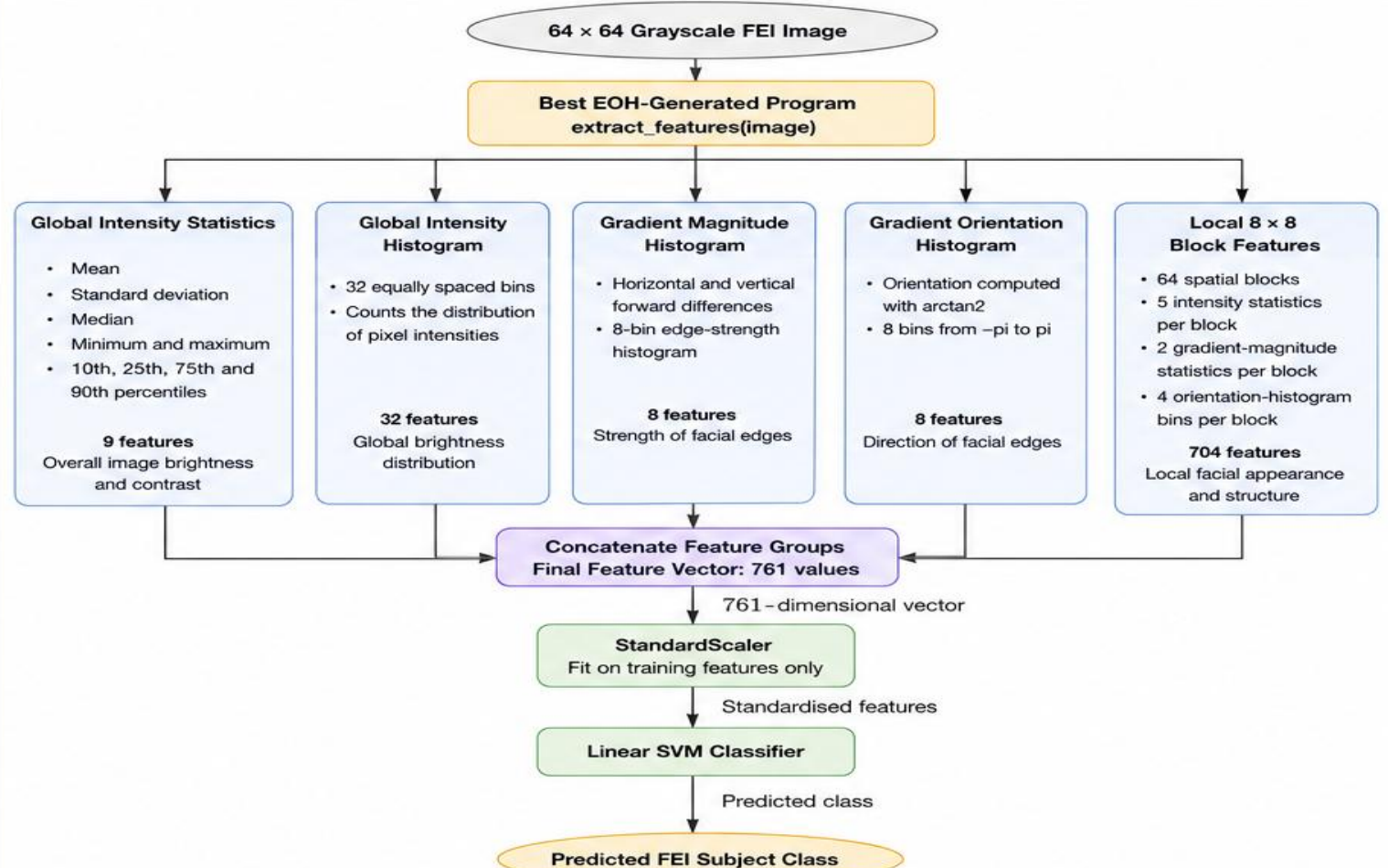


Together, these groups produce a rich and interpretable representation of facial images, combining **global appearance**, **edges** and **local facial information**.



**Key Takeaway:** The EOH-generated feature extractor transforms a  $64 \times 64$  grayscale image into a 761-dimensional feature vector using **explicit, mathematically defined** operations. Every feature corresponds to an observable image property, ensuring high **transparency** and **interpretability**.

## Operational Pipeline of the Best EOH-Generated FEI Feature Extractor





# 6. Why is the LLM Feature Extractor Explainable?



This LLM-generated feature extractor is highly explainable because every feature is derived from **explicit, well-defined mathematical operations** that correspond to **observable image properties**.

| Why This Approach is Explainable                                                   |                                                                                                                                              |
|------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
|    | <b>Transparent and Interpretable</b><br>Every feature has a clear mathematical definition and an intuitive meaning.                          |
|    | <b>Traceable to Image Properties</b><br>Features correspond to observable properties such as brightness, edges, texture and local structure. |
|    | <b>Easy to Verify and Debug</b><br>Each feature can be independently inspected, verified and validated.                                      |
|    | <b>Low Black-Box Behaviour</b><br>The extractor uses no hidden layers or latent representations—only explicit computations.                  |
|  | <b>Balance of Power and Interpretability</b><br>Captures rich global, edge and local information while remaining fully transparent.          |

| Explainability Assessment                                                                                                |                    |                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------|--------------------|------------------------------------------------------------------------------------|
| Property                                                                                                                 | Assessment         | Justification                                                                      |
|  <b>Mathematical Interpretability</b> | ★★★★★<br>Excellent | All features are computed using explicit, well-defined mathematical operations.    |
|  <b>Human Interpretability</b>        | ★★★★★<br>Excellent | Each feature maps to an intuitive and observable image property.                   |
|  <b>Transparency</b>                  | ★★★★★<br>Excellent | The full process from image to feature vector is fully visible and understandable. |
|  <b>Traceability</b>                  | ★★★★★<br>Excellent | Every feature can be traced back to a specific computation and image region.       |
|  <b>Ease of Debugging</b>            | ★★★★★<br>Excellent | Individual features can be inspected and tested independently.                     |
|  <b>Black-Box Behaviour</b>         | ★☆☆☆☆<br>Very Low  | No hidden layers or latent variables—only explicit and interpretable computations. |



**Overall Explainability Rating:** **High (5/5)**

The LLM-generated feature extractor achieves an excellent level of explainability, making it a **transparent** and **trustworthy** solution for explainable machine learning in facial image classification.

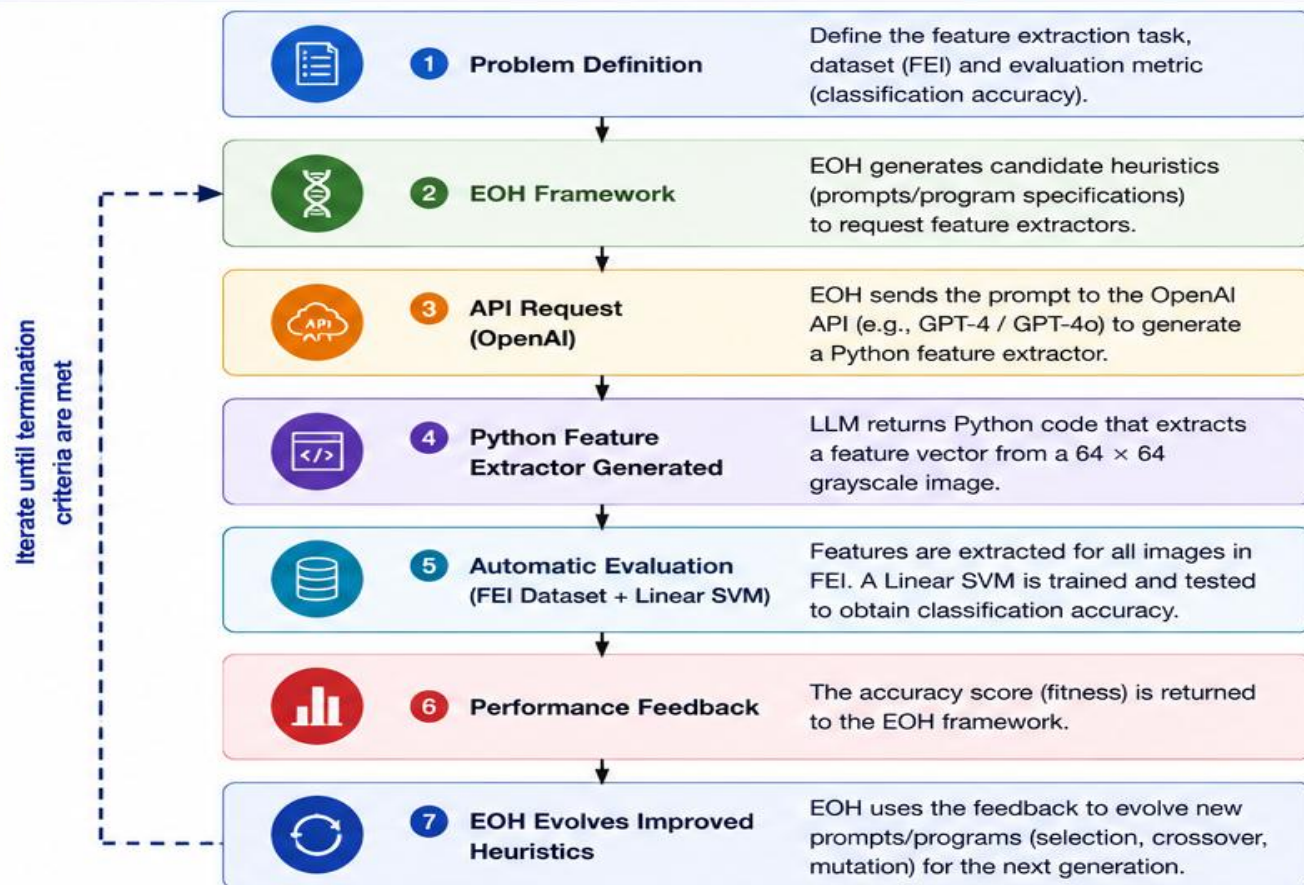


# 7. Developing the LLM Feature Generation Pipeline



This week I developed a complete pipeline that uses the **EOH framework** and an **LLM** (via **OpenAI API**) to automatically **generate**, **evaluate** and **evolve** feature extraction algorithms for facial image classification.

## Pipeline Overview



## Key Components



### OpenAI API

- Secure API key used to access GPT model.
- Generates Python code for feature extraction.



### EOH Framework

- Evolves prompts/program specifications.
- Uses evolutionary operators to search the space of possible heuristics.



### FEI Dataset

- $64 \times 64$  grayscale facial images.
- Used for feature extraction and evaluation.



### Evaluation (Linear SVM)

- Linear SVM trained on extracted features.
- Accuracy used as the fitness score.



### Feedback Loop

- Performance guides EOH to generate better feature extractors.
- Enables an automated evolutionary search.



### Goal of the Pipeline

To automatically discover high-performing, explainable feature extraction algorithms by combining the creativity of LLMs with the optimisation power of EOH.

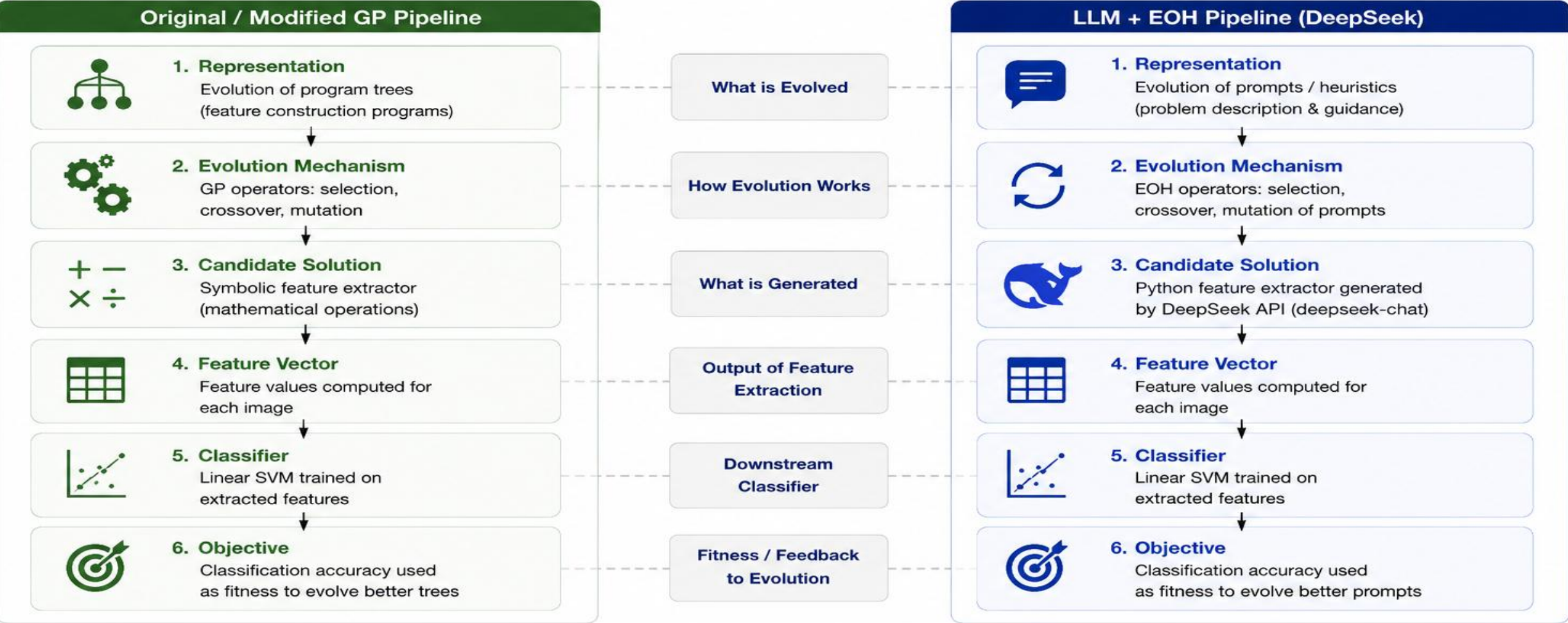


**Key Takeaway:** This pipeline allows Large Language Models to go beyond one-off code generation and become part of an **evolutionary optimisation framework**, iteratively discovering and improving feature extraction algorithms.



# 8. GP vs LLM: Pipeline Comparison

 Both approaches aim to automatically discover explainable feature extraction algorithms for facial image classification. **They follow different routes but share the same end goal.**



 **Key Takeaway:** GP evolves mathematical programs using genetic operators, while LLM + EOH evolves natural language prompts that generate Python code via the DeepSeek API. Both produce a feature vector for the same classification task and are **evaluated identically.**



# 9. GP vs LLM: Explainability Comparison



Both GP and the LLM-generated feature extractor produce explicit, interpretable features.

The LLM approach offers higher human interpretability and easier modification of the extracted features.

| Criterion                                                                                                                                                                      | Genetic Programming (Original / Modified GP)                                                       | LLM + EOH (Generated Feature Extractor)                                                                                     | Comparison                                                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <b>Transparent Operations</b><br>Are operations explicit and interpretable?                   | ★★★★★<br>Every node in the tree is an explicit mathematical operation.                             | ★★★★★<br>All features are computed using explicit Python operations (statistics, histograms, gradients, local block stats). |  <b>Comparable</b><br>Both are fully transparent with no hidden layers.                      |
|  <b>Human Readability</b><br>How easy is it for a human to understand?                         | ★★★★☆<br>Tree structures and primitive combinations can be complex to interpret.                   | ★★★★★<br>Python code with meaningful variable names and clear operations is easy for humans to read.                        |  <b>LLM Advantage</b><br>Code is more intuitive and easier to follow for most users.         |
|  <b>Mathematical Interpretation</b><br>Do features have clear mathematical meaning?            | ★★★★★<br>Each tree node corresponds to a well-defined mathematical function.                       | ★★★★★<br>Each feature maps to a specific and well-understood image property (e.g., mean intensity, gradient magnitude).     |  <b>Comparable</b><br>Both approaches yield features with clear mathematical interpretation. |
|  <b>Ease of Debugging and Modification</b><br>How easily can features be inspected or changed? | ★★★★☆<br>Trees can be inspected but are less intuitive to modify; changes require GP re-evolution. | ★★★★★<br>Python code is easy to inspect, debug and modify directly by editing the operations or parameters.                 |  <b>LLM Advantage</b><br>Much easier to debug and adapt the feature extractor.               |
|  <b>Hidden Representations</b><br>Are there latent or opaque representations?                | ★★★★★<br>No hidden representations; everything is explicit in the tree.                            | ★★★★★<br>No hidden representations; features are explicitly computed from the image.                                        |  <b>Comparable</b><br>Both are fully interpretable with no latent variables.               |



**Overall Explainability Strengths**

- Both methods generate highly transparent and mathematically grounded features.
- LLM-generated features are more human-readable and easier to modify.



**Key Takeaway**

LLM + EOH provides comparable mathematical interpretability to GP, while offering superior human interpretability and development agility.



Both approaches produce explainable features, but the LLM-generated extractor is easier to understand, modify and extend in practice.



# 10. GP vs LLM: Performance Comparison (Current Status)



We compare Genetic Programming (GP) approaches with the new LLM + EOH (DeepSeek) pipeline.  
GP results are taken from the literature. LLM results are currently under evaluation.

| Metric                                                                                                                | Original GP                                                           | Modified GP                                                         | LLM + EOH (DeepSeek)                                                   |
|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------|------------------------------------------------------------------------|
|  Mean Accuracy                        | <b>96.83%</b><br>(from paper)                                         | <b>96.58%</b><br>(from paper)                                       | <b>Evaluation in Progress</b><br>(to be determined)                    |
|  Best Accuracy                        | <b>97.56%</b><br>(from paper)                                         | <b>100.00%</b><br>(from paper)                                      | <b>Evaluation in Progress</b><br>(to be determined)                    |
|  Explainability (Transparency)        | <b>High</b><br>Explicit mathematical operations in trees              | <b>High</b><br>Explicit mathematical operations in trees            | <b>Very High</b><br>Python code with clear, interpretable operations   |
|  Development Time                    | <b>High</b><br>Population evolution is computationally intensive      | <b>High</b><br>Additional domain knowledge and tuning required      | <b>Low</b><br>Rapid code generation via LLM; minimal manual design     |
|  Automation (Level of Human Design) | <b>Low</b><br>Manual design of function set, operators and parameters | <b>Low</b><br>Requires handcrafted modifications and domain insight | <b>High</b><br>Prompts + EOH handle iterative refinement automatically |

| Current Findings                                                                      |                                                                                                                                                                     |
|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <b>GP Provides Strong Validated Performance</b><br>Original and Modified GP achieve very high classification accuracy (up to 100% best accuracy in the literature). |
|    | <b>LLM Generates Explainable Algorithms</b><br>The LLM (DeepSeek) successfully produces fully interpretable feature extractors expressed as Python code.            |
|    | <b>LLM Reduces Development Effort</b><br>LLM + EOH pipeline dramatically reduces the time and domain expertise required to design feature extraction algorithms.    |
|  | <b>Performance Evaluation Ongoing</b><br>LLM-generated extractors are being evaluated on the FEI dataset. Results will be available in the next phase.              |



**Key Takeaway:** GP approaches currently deliver the highest validated accuracy, while the LLM + EOH pipeline offers greater automation, explainability, and efficiency. Performance comparison for LLM-generated extractors is in progress.